

10 period of sound wave, T : $T = \frac{1}{f}$

$$\theta = \frac{2\pi}{16} = \frac{\pi}{8} \text{ rad}$$

$$\omega = \frac{\theta}{T} = \theta f = \left(\frac{\pi}{8}\right)(2200) = 864 \text{ rad s}^{-1}$$

11 gravitational force provides centripetal force